Mass (m) - is **the amount of matter or substance that makes up an object**. It is measured in kilograms, denoted as kg Mass always remains the same

Length is a measure of distance. Height of tank (h) length of pipe L, width of a tank b,w Diameter of pipe D

Time- the measured or measurable period during which an action, process, or condition exists measured in seconds , minutes, hours denoted as s,min,hr some refer seconds as sec

Volume- **Volume** is the quantity enclosed by a closed surface
It is measured in litres, m^3

Velocity -Velocity is **the speed of an object moving in a definite direction**. The SI unit of velocity is also metre per second. Velocity is a vector quantity; it has both magnitude and direction. Denoted as u,v and has dimension of m/s

Density-of a substance is its mass per unit volume. The symbol used for density is ρ =m/V

Acceleration is the rate of change of the velocity of an object with respect to time. Denoted as a, dimension- m/s^2
Acceleration due to gravity is g

Relative density/Specific gravity-is the ratio of the **density** (mass of a unit volume) of a substance to the **density** of a given reference material. No dimension but can be used to find density of fluid.

Eg specific gravity of 0il is 0.8 means density is 800 kg/m^3

Force: is an influence that can change the motion of an object.

F = ma, It is measured in the SI unit of Newton (N)

A Newton is defined as 1 kg·m/s^2, which is the force which gives a mass of 1 kilogram an acceleration of 1 metre per second, per second.

Pressure is referred in Force/area and also in terms of a column of liquid

Dimension N/m^2 , 760 mmHg,

Work done *W* by a constant force of magnitude *F* on a point that moves a displacement *s* in a straight line in the direction of the force is Force*distance. Dimension Nm

Power refers "rate of doing work" pumping **power**, Nm/s, watts, Hp

Volumetric flow rate /Discharge from a pipe or tank, referred as Q, Dimension litres/s, m^3/s

Mass flow rate referred as $\dot{m}$, Dimensions kg/s

Viscosity: The **viscosity** of a fluid is a measure of its resistance to deformation at a given rate.

In Momentum transfer conceptualized as the internal frictional force that arises between adjacent layers of fluid that are in relative motion. referred as μ

Dimension kg/ms,or Ns/m^2

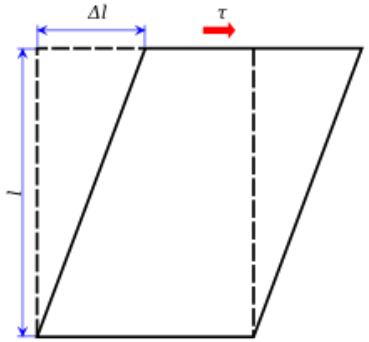
Kinematic viscosity

In Momentum transfer , it is sometimes more convenient to work in terms of the *kinematic viscosity* also called the *momentum diffusivity*, defined as the ratio of the viscosity *μ* to the density of the fluid *ρ*. It is usually denoted by nu $\nu = \mu/\rho$ *Dimension* m^2/sec

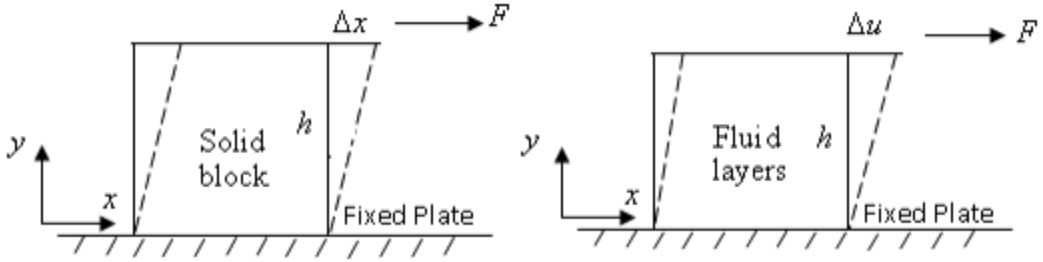
specific weight, also known as the **unit weight**, is the weight per unit volume of a material. The specific weight, *γ*, of a material is defined as the product of its density, *ρ*, and the gravity, *g*

$\gamma = \rho g$ Dimension kg/m^2s^2

Shear stress, often denoted by $\boldsymbol{\tau}$ (is the component of stress coplanar with a material cross section



The shear modulus of solid (S) and coefficient of viscosity (μ) for fluid can defined as



$$S = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{(F/A)}{(\Delta x/h)}; \quad \mu = \frac{\text{Shear stress}}{\text{Shear strain rate}} = \frac{(F/A)}{(\Delta u/h)}$$

Ideal gas law

Mathematically is expressed as;

$PV=nRT$

Where,

V = volume of gas.m^3

T = temperature of the gas.K

P = pressure of the gas.N/m^2

R = universal gas constant8314J/kgmoleK

n denotes the number of moles

By replacing n with m/M and subsequently

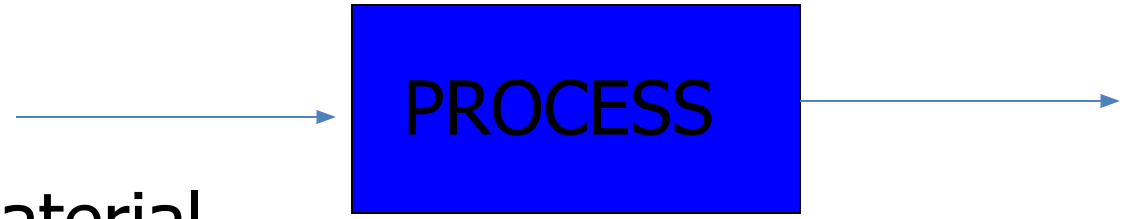
total mass of the gas (m) (in kilograms) divided by the molar mass or molecular weight (M) (in kilograms per mole)

introducing density $\rho = m/V$, we get

$P=PM/RT$

Introduction to Chem.Engg.

Typical process

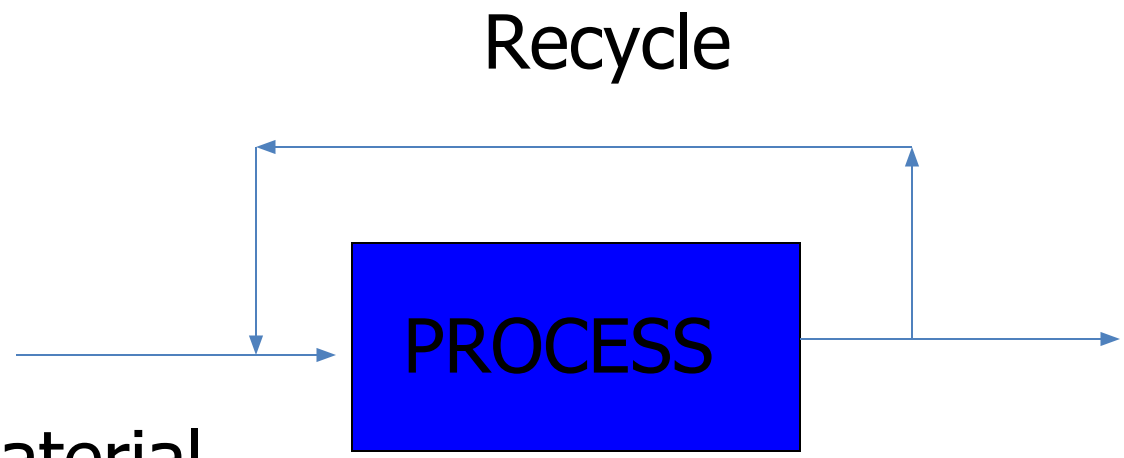


Ram material preparation

Upstream processing

Product purification

Downstream processing



Ram material preparation

Product purification

Upstream processing

Downstream processing

- Unit operation accounts for 90% of the cost of a plant.
  >Processing equipment is a Reactor
  > **Reactor design** uses information, knowledge and experience from a variety of areas –Thermodynamics, Kinetics,Fluid mechanics,Heat transfer,Mass transfer, Material science and Economics.

>Raw material preparation –Mecahnical Operations(Size reduction,filtration sedimentation)
Fluid mechanics (fluid flow, pumps etc.,)
Heat transfer _(STHE,DPHE,Condensor etc.,)
Mass transfer equipments (Evaporation, Humidification etc)

>Product purification-M.O, MoT(Fluid mechanics),H.T, Mass Transfer (Evaporation,Distillation, Crystallization, Drying, etc.,
>For efficient processing : Process control and Simulation,
Computation,Modelling and Scaleup

MOMENTUM TRANSFER

Course Overview*:*
Understanding the behavior of fluids is important for accurately treating problems on the movement of fluid through pipes, pumps and all kinds of process equipment.

This subject will pave way for students to learn the behavior of various types of fluids and to find practical solutions to processing problems.

Key take away from the Course

- Explain variations of pressure in static fluids
- Classify various types of fluid and explain its flow behavior
- Understand the nature of fluid flow in different conduits, packed columns and open channels
- Predict time of emptying tanks
- Measure flow rates using appropriate measuring instruments
- Explain the application and functions of pumps
- Obtain functional relationships using dimensional analysis

Brief Course outline:

- Introduction to Fluid Static's and dynamics
- Concepts of pressure, density and viscosity
- Manometry,
- Decanters
- Fluid flow phenomena
- Basic equations of fluid flow
- Flow measurement
- Velocity profiles in laminar and turbulent fluid
- Pumps and fluid moving machineries
- Unsteady fluid flow
- Dimensional analysis
- Flow of compressible fluids.

Momentum transfer

- What is momentum Transfer?
- What makes fluid flow?
- Two paper boats placed in a rivulet ,one at the centre and one near the bank?
- Two bottles one filled with honey and one filled with water is tilted?
- When an iron ball is dropped in a deep sea what happens to its velocity?





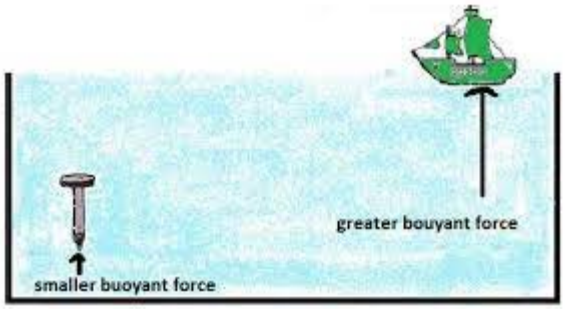




- When you draw water from a well you feel light when the bucket is inside water ?
- Why a piece of wood floats and piece of iron sinks?



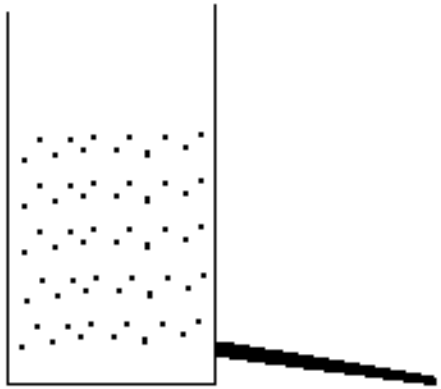


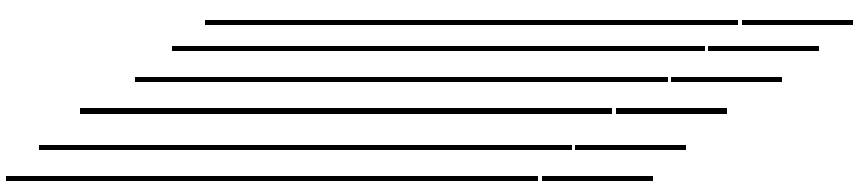


Answers

- Fluid – any thing which flows liqid or gas
- Momentum- Mass X Velocity

Fluid flows due to---





Deck of cards



Course Content
Course Code: 18CH34
Course Title: Momentum Transfer **CIE: 100**
Teaching Hours: Th-22 **SEE: 100**
Part A

UNIT-I

FLUID STATICS AND ITS APPLICATIONS: Introduction to Unit operations, Concept of Momentum Transfer, Nature of fluids and pressure concept, Variation of pressure with height –hydrostatic equilibrium, Barometric equation, Measurement of fluid pressure – manometers. Decanter, Continuous gravity decanter, Centrifugal decanter. **FLUID FLOW PHENOMENA:** Types of fluids – shear stress and velocity gradient relation, Newtonian and non – Newtonian fluids, Viscosity of gases and liquids. Types of flow – laminar and turbulent flow, Reynolds stress, Eddy viscosity. Flow in boundary layers, Reynolds number, Boundary layer separation and wake formation. 11hrs

UNIT-II

BASIC EQUATIONS OF FLUID FLOW: Average velocity, Mass velocity, Continuity equation, Euler and Bernoulli equations, Modified equations for real fluids with correction factors. Pump work in Bernoulli equation. Angular momentum equation. **FLOW OF INCOMPRESSIBLE FLUIDS IN CONDUITS AND THIN LAYER**: Laminar flow through circular and non-circular conduits, Hagen Poiseuille equation, Laminar flow of non-newtonian liquids. Turbulent flow in pipes and closed channels, Friction factor chart, Friction due to change in velocity or direction. Friction losses in Bernoulli equation. 10Hrs

UNIT-III

Flow of Fluids Past Immersed Bodies: Pressure drop studies in packed bed –Ergun, Kozeny-Carman and Blake-Plummer Equations, Fluidization, Conditions for fluidization, Minimum fluidization velocity, Types of fluidization, Applications of fluidization, Slurry transport, Pneumatic conveying

INTRODUCTION TO UNSTEADY STATE FLOW: Time to empty the liquid from a tank- , Rectangular, Cylindrical (Horizontal and Vertical), Hemi spherical. 10Hrs

UNIT-IV

TRANSPORTATION AND METERING OF FLUIDS: Pipes, Fitting and valves, Measurement of liquid and gas flow rates by Pitot tube, Orifice meter, Venturi meter and Rota meter. Flow through open channels – weirs and notches. Performance and characteristics of pumps – Positive displacement and centrifugal pumps.Fans,Compressor and Blowers.10Hrs

UNIT-V

DIMENSIONAL ANALYSIS: Dimensional homogeneity, Rayleigh's and Buckingham's II – methods. Significance of different dimensionless numbers. Elementary treatment of similitude between model and prototype.

FLOW OF COMPRESSIBLE FLUIDS: Continuity equation, Concept of Mach number, Total energy balance, Velocity of sound, Ideal gas equations. Flow through variable-area conduits. Adiabatic frictional flow. Isothermal frictional flow (elementary treatment only) 11Hrs

Reference Books:

McCabe and Smith W.L., “Unit Operations of Chemical Engineering”, 7th Edition, McGraw Hill, New York, 2007.

Coulson J. M., and Richardson. J.F., “Chemical Engineering” Vol. 2., 5th Edition, Asian Books (p) Ltd., New Delhi, 2003

Badger W.I., and Banchero J.T., “Introduction to Chemical Engineering”, 7th Edition, Tata McGraw Hill, New York, 2007.

Kumar K. I. “ Engineering Fluid Mechanics”,3rd Edition Eurasia Publishing House (p) Ltd., New Delhi., 2009.

Course Outcomes:

Upon successful completion of the course students will be able to

Co1 Recall the concepts of fluid statics and dynamics.

Co2 Explain the fundamental equations of fluid flow

Co3 Analyze the flow behavior in various geometries and packed columns

Co4 Apply fluid flow principles in flow measurement, power required for transportation and energy losses.

CIE is executed by way of quizzes (Q), tests (T) and experiential learning (EL). A minimum of three quizzes are conducted and each quiz is evaluated for 10 marks adding up to 30 marks. All quizzes are conducted online. Faculty may adopt innovative methods for conducting quizzes effectively. The number of quizzes may be more than three also. The three tests are conducted for 50 marks each and the sum of the marks scored from three tests is reduced to 50. The marks component for experiential learning is 20.

Total CIE is 30(Q) +50(T) +20(EL) =100 Marks.